



Time = 20 Minutes

Quiz No 08

[20 points]

Provide all your answers on the answer sheet. Present your final answers and reasoning clearly. Most of the total score for the questions will be awarded on your reasoning.

CLO No 04 - Construct probability distributions, up to two dimensions, from given data for discrete and continuous random variables. (Cog - 3)

Question 1 [14 points] [Joint Probability Density Function, 2-dimensional]

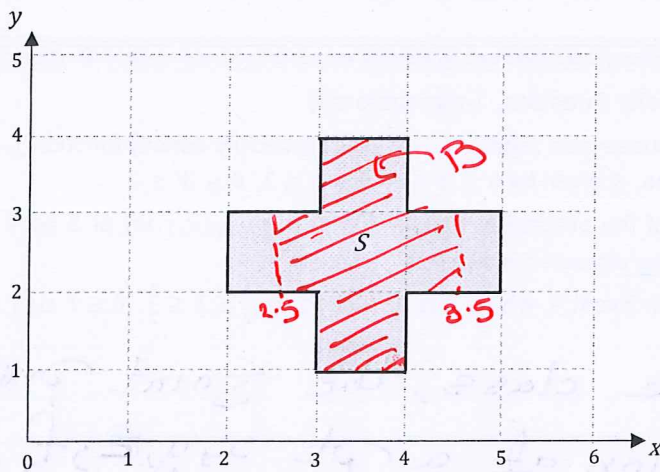
Random Variable X and Y are jointly continuous on the shaded region S (see below). The joint probability density function $f_{X,Y}(x,y)$ is uniformly distributed over the region S . **Note** - provide necessary justification for all the following parts.

Part a [02 points] Determine the joint probability density function $f_{X,Y}(x,y)$.

Part b [02 points] Calculate the probability of Event A, where A is described where random variable $Y = X$.

Part c [02 points] Calculate the probability of Event B, where B is given by $2.5 < X < 4.5$ and $1 < Y < 4$.

Part d [08 points] Calculate and plot $f_X(x)$ and $f_Y(y)$.



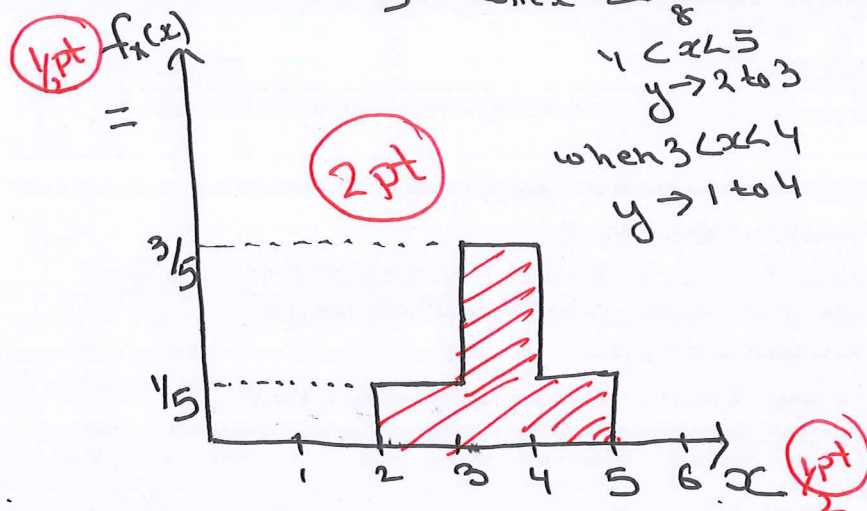
(a) Area of $S = 5 \text{ unit}^2 \Rightarrow f_{X,Y}(x,y) = \begin{cases} 1/5 & \text{within } S \\ 0 & \text{else} \end{cases}$ 2pts

(b) Event A, $Y = X$ is a straight line whose Area = 0 $\Rightarrow P(A) = 0$ 2pts
with no Area $\Rightarrow$ No Volume.

(c) $P(B) = 1/5 \times \text{Area spanned by } 2.5 < X < 4.5 \text{ and } 1 < Y < 4 \text{ within } S$
 $= 1/5 \cdot \left[\frac{1}{2} \cdot 1 + 1 \cdot 3 + \frac{1}{2} \cdot 1 \right]$ 1pt
 $= 4/5$ 1pt

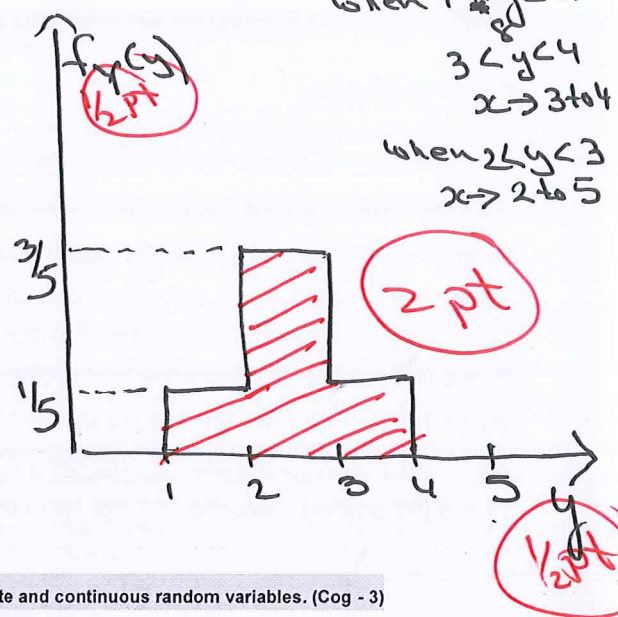
$$(d) f_X(x) = \int f_{X,Y}(x,y) dy$$

when $2 < x < 3$
 $1 < x < 5$
 $y \rightarrow 2 \text{ to } 3$
 when $3 < x < 4$
 $y \rightarrow 1 \text{ to } 4$



$$f_Y(y) = \int f_{X,Y}(x,y) dx$$

when $1 < y < 2$
 $3 < y < 4$
 $x \rightarrow 3 \text{ to } 4$
 when $2 < y < 3$
 $x \rightarrow 2 \text{ to } 5$



CLO No 04 - Construct probability distributions, up to two dimensions, from given data for discrete and continuous random variables. (Cog - 3)

Question 2 [06 points] [Joint Probability Density Function, 3-dimensional]

Random Variable X , Y and W are jointly continuous on a region S . The joint probability density function $f_{X,Y,W}(x, y, w)$ is uniformly distributed over the 3-dimensional cube, S , given by $0 \leq X \leq 2$, $0 \leq Y \leq 2$, $0 \leq W \leq 2$.

Part a [03 points] What is the interpretation of the probability density function, $f_{X,Y,W}(x, y, w)$ at a point (x, y, w) in S .

Note - make sure the interpretation explains everything discussed in the class.

Part b [03 points] Calculate the probability of an Event A, where A is described by, $\frac{1}{2} \leq X \leq \frac{3}{2}$, $0 \leq Y \leq \frac{3}{2}$, $0 \leq W \leq 2$.

(a) As discussed in the class, the joint probability density function at a pt x, y, w , $f_{X,Y,W}(x, y, w)$ is the probability density (Probability per unit Volume) at x, y, w . 1 pt

$$\Rightarrow P(x \leq X \leq x + \Delta, y \leq Y \leq y + \Delta, w \leq W \leq w + \Delta) \approx f_{X,Y,W}(x, y, w) \cdot \Delta^3$$

2 pt

(b) Region in A is defined by $\frac{1}{2} \leq X \leq \frac{3}{2}$, $0 \leq Y \leq \frac{3}{2}$, $0 \leq W \leq 2$. So $\frac{1}{2} \leq X \leq \frac{3}{2}$, $0 \leq Y \leq \frac{3}{2}$, $0 \leq W \leq 2$

$$\text{Volume of } S_0 \text{ in } S = \left(\frac{3}{2} - \frac{1}{2}\right) \left(\frac{3}{2}\right) (2) = 1 \times \frac{3}{2} \times 2 = 3 \text{ unit}^3$$

1 pt

$$f_{X,Y,W}(x, y, w) = \begin{cases} 1/8 & \text{in } S \\ 0 & \text{else} \end{cases}$$

1 pt

$$\text{Therefore, } P(A) = 3 \times 1/8 = 3/8$$

1 pt